

ASSIGNMENT 4.1

❖ Firewall Basics Theory:

Task (1):

Blocking Spotify with an outbound rule:

The screenshot shows the 'New Outbound Rule Wizard' window in Windows Firewall. The title bar reads 'New Outbound Rule Wizard' with a close button. The main heading is 'Program'. Below it, the instruction says: 'Specify the full program path and executable name of the program that this rule matches.'

On the left, a 'Steps:' pane lists five steps: 'Rule Type' (highlighted in blue), 'Program' (highlighted in green), 'Action', 'Profile', and 'Name'.

The main area contains the question: 'Does this rule apply to all programs or a specific program?'. There are two radio button options:

- ☐ **All programs**
Rule applies to all connections on the computer that match other rule properties.
- ☒ **This program path:**
Below this option is a text input field containing '%USERPROFILE%\Downloads\Supremo.exe' and a 'Browse...' button to its right. Below the input field, there are example paths: 'Example: c:\path\program.exe' and '%ProgramFiles%\browser\browser.exe'.

At the bottom right, there are three buttons: '< Back', 'Next >', and 'Cancel'.



Action

Specify the action to be taken when a connection matches the conditions specified in the rule.

Steps:

- Rule Type
- Program
- **Action**
- Profile
- Name

What action should be taken when a connection matches the specified conditions?

☐ **Allow the connection**

This includes connections that are protected with IPsec as well as those are not.

☐ **Allow the connection if it is secure**

This includes only connections that have been authenticated by using IPsec. Connections will be secured using the settings in IPsec properties and rules in the Connection Security Rule node.

[Customize...](#)

☒ **Block the connection**

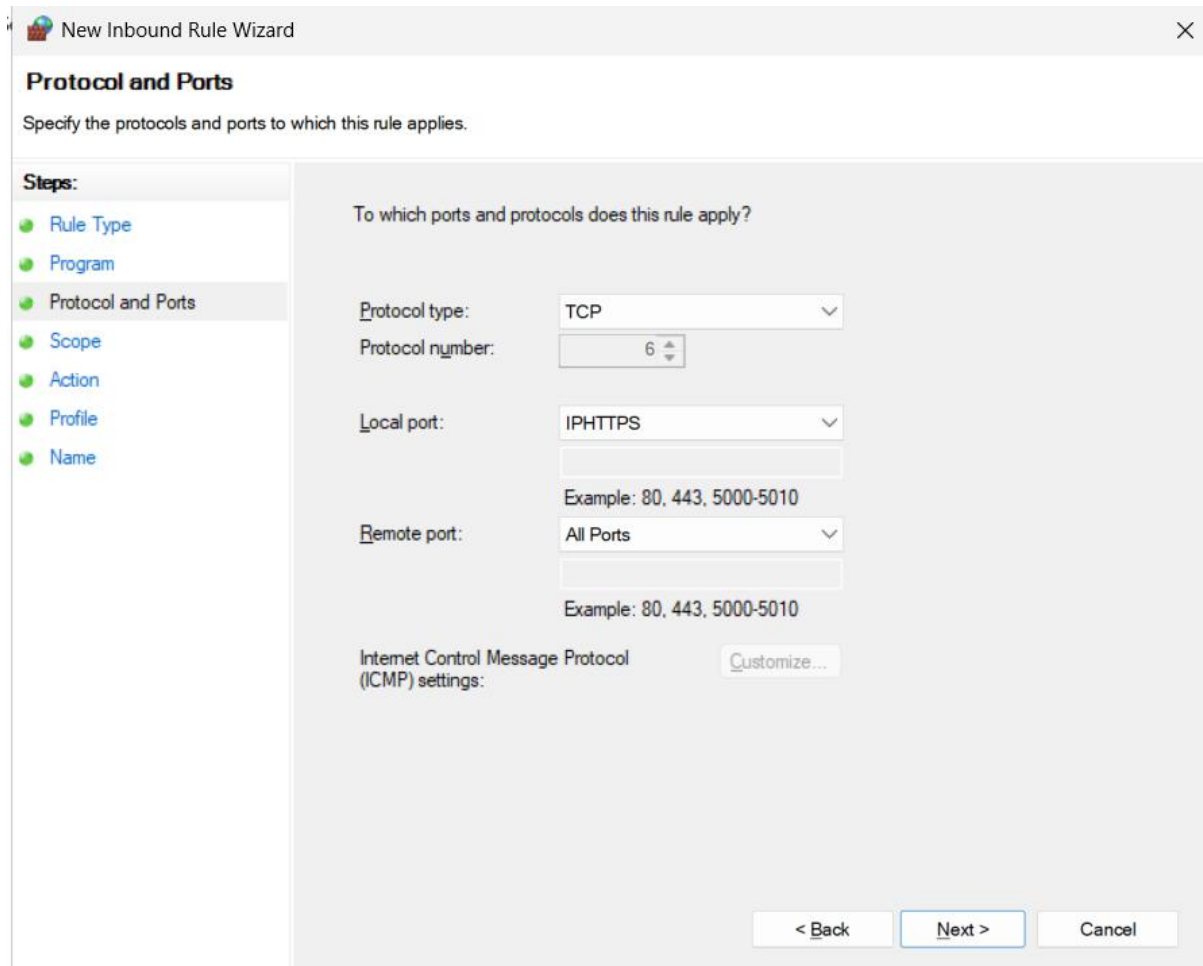
< Back

Next >

Cancel

Task (2):

Allowing Chrome inbound only on port 443:



The image shows the 'New Inbound Rule Wizard' window, specifically the 'Protocol and Ports' step. The window title is 'New Inbound Rule Wizard' with a close button. The subtitle is 'Protocol and Ports' and the instruction is 'Specify the protocols and ports to which this rule applies.' On the left, a 'Steps:' sidebar lists: Rule Type, Program, Protocol and Ports (selected), Scope, Action, Profile, and Name. The main area is titled 'To which ports and protocols does this rule apply?'. It contains the following fields: 'Protocol type:' set to 'TCP'; 'Protocol number:' set to '6'; 'Local port:' set to 'IPHTTPS' with an example '80, 443, 5000-5010'; 'Remote port:' set to 'All Ports' with an example '80, 443, 5000-5010'; and 'Internet Control Message Protocol (ICMP) settings:' with a 'Customize...' button. At the bottom right are '< Back', 'Next >', and 'Cancel' buttons.

New Inbound Rule Wizard

Protocol and Ports

Specify the protocols and ports to which this rule applies.

Steps:

- Rule Type
- Program
- Protocol and Ports**
- Scope
- Action
- Profile
- Name

To which ports and protocols does this rule apply?

Protocol type: TCP

Protocol number: 6

Local port: IPHTTPS

Example: 80, 443, 5000-5010

Remote port: All Ports

Example: 80, 443, 5000-5010

Internet Control Message Protocol (ICMP) settings: Customize...

< Back Next > Cancel

Task (3):

Example IP address: 123.45.67.89

1. Open Windows Defender Firewall

- 1) Press Windows + R.
- 2) Type:
- 3) wf.msc
- 4) Press Enter.
- 5) The Windows Defender Firewall with Advanced Security window will open.

2. Create a New Inbound Rule

- 1) In the left panel, click Inbound Rules.
- 2) In the right Actions panel, click New Rule...
- 3) Select Custom and click Next.

3. Select Protocol and Ports

- 1) Keep the default settings:
 - Protocol type: Any
 - Local port: All Ports
 - Remote port: All Ports
- 2) Click Next.

4. Select Protocol and Ports

- 1) Keep the default settings:
 - Protocol type: Any
 - Local port: All Ports
 - Remote port: All Ports

2)Click Next.

5.Specify the IP Address — Scope Tab

- 1.Under Which remote IP addresses does this rule apply to, select:
- 2.These IP addresses
- 3.Click Add...
- 4.Enter:123.45.67.89
- 5.Click OK.
- 6.Click Next.

Task (4):

Hardware Firewall vs. Software Firewall:

Difference	Hardware Firewall	Software Firewall
1. Location	A separate physical device placed between the network and the internet.	Installed and runs directly on a computer or server.
2. Protection	Protects multiple devices on an entire network.	Mainly protects the individual device where it is installed.
3. Performance & Management	Designed for high-volume traffic and centralized network management.	Uses the computer's resources and is managed on individual systems.